

MC904

v 1.0

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Power By MettaX

Version Control				
Version	Date	Content	Modified By	Approver
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1. Basic Parameters

Operating System	Linux
CPU	ARM Cortex CA9 Dual Core + NPU@0.7
RAM	DDR 512 Mb
Memory Card	2*8G~512G TF Card
G-Sensor	3 Axis Acceleration Sensor
Speaker	External speaker
Mic	External MIC
Sim card	1 Standard SIM card port
LED	3 LED light , Power, Recording, Networking
USB	1 USB port, support mouse control
IR	Support infrared remote control
I/O Ports	4 digital inputs,2 RS232
Power	Adaptive wide range of power input, with overload, undervoltage, short-circuit, reverse connection and other protection functions; support for timer on/off, delayed shutdown function.
Input Voltage	9-36V
Power Wastage	1.42A
Operation Temperature	-20℃ to +70℃
Operation Humidity	8% to 90%
Dimensions	120*120*46mm

2. Cellular

2.1. Bands

MC904-EU:

FDD LTE: B1/B3/B5/B7/B8/B20/B28

TDD LTE: B38/B40/B41

WCDMA: B1/B5/B8

GSM: 850/900/1800/1900MHz

MC904-LA:

FDD LTE: B2/B3/B4/B5/B7/B12/B18

TDD LTE: B38/B41

WCDMA: B2/B4/B5

GSM: 850/900/1800/1900MHz

MC904-NA:

LTE FDD: B2/4/5/12/13/14/66/71

WCDMA: B2/4/5

2.2. Wi-Fi

Bands : 2.4GHz, 802.11 b/g/n

Usage : AP,Station

3. Camera

3.1. Video recording parameters

Video Standards	PAL, NTSC
Video Compression Format	H.264/H.265 compression format
Video Resolution	Supports 1080P, 720P, D1 optional, total encoding resources 1080P 60 frames
Video Quality	Image quality from 1 to 8, with the best level 1 and the lowest level 8
Recording Rate	CIF: 256Kbps ~ 1.5Mbps, multi-level image quality optional HD1: 600Kbps ~ 2.5Mbps, multi-level image quality optional D1: 800Kbps ~ 3Mbps, multiple quality levels available 720P: 1Mbps-4Mbps, multi-level image quality optional 1080P: 1Mbps~8Mbps, multi-level image quality optional
Recording Method	Automatic recording by default, supports ignition recording, alarm recording, etc.
Audio Input	Support 4-way camera audio input, sound and video recording synchronization
Audio Output	Supports external speaker output
Compression Format	G.711A compression format
Recording Method	Simultaneous recording of sound and video

4. GNSS

GPS Module	External Gmouse, TTL Level Interface
Positioning Frequency	Default 20 seconds Minimum 1 second Maximum 600 seconds

5.AI

5.1. ADAS

Alarm Type	Detection Source	Triggering Conditions
Forward Collision	Is there any risk of collision between the vehicle and the vehicle in front?	Real-time monitoring of the distance between the vehicle and the vehicle in front. When the TTC is less than 2 seconds (default is 2s, configurable (configured during calibration)), the system issues a warning
Lane Departure	Whether the vehicle deviates from the current lane	It is judged by the lane line recognition algorithm that when the outer edge of the wheel approaches/crosses the lane line a certain distance (configurable, the range is plus or minus 20cm, the default value is -20cm, that is, it crosses the lane line 20cm)
Vehicles Too Close	The distance between this vehicle and the vehicle in front is less than the minimum safe distance	Real-time monitoring of the distance between the vehicle in front and the vehicle in front. When the TTC is less than 0.8 seconds (default value, configurable (configured during calibration)), the system issues a warning
Pedestrian Collision	Are there any potential collision risks between vehicles and pedestrians?	Pedestrian anti-collision warning, pedestrian recognition through the camera, when the TTC is less than 2 seconds (default value, configurable (configuration during calibration)), the alarm is started

5.2. DMS

Alarm Type	Detection Source	Triggering Conditions
Drowsy driving	Is the driver closing his eyes or yawning?	(Eyes and mouth) If the eyes are not covered, squinting or closing the eyes (the eyelids are closed more than 90%) or opening the mouth widely (the mouth is open more than 90%) for more than 2 seconds (configurable), an alarm will be triggered.
Make phone calls and call the police	When the vehicle is in a driving state, it can conduct comprehensive identification and analysis based on the handheld phone items and the actions of making and receiving calls.	If the driver holds the phone to the ear or mouth at a minimum distance of no more than 5cm, and the behavior lasts for more than 2 seconds (configurable), the alarm will be triggered.
Smoking alarm	Detect whether the driver is holding a cigarette in his mouth or holding a cigarette stick	The driver holds a cigarette no larger than 5cm from his mouth or holds a cigarette in his mouth. When it lasts for more than 2s (configurable), an alarm is triggered.
Distracted driving	When the vehicle is in a driving state, it can comprehensively recognize and analyze facial gestures such as looking left and right, looking up, and lowering your head.	The alarm is triggered when the face posture is greater than or equal to 25 degrees (configurable) to the left or right, or greater than or equal to 20 degrees (configurable) up and down relative to the front, and lasts for more than 2 seconds (configurable).
Driver abnormality	When the vehicle is in a driving state, comprehensive identification and analysis of the driver's absence from the driving position or deviation from the driving position, etc.	When no face is detected or the face is far away from the center of the screen for more than 2 seconds (configurable), an alarm is triggered.
The camera is blocked	Detect whether the camera is blocked through image recognition analysis	If it is detected that the camera is blocked for more than 2 seconds (configurable), an alarm will be triggered.

5.3. DMSX

Alarm Type	Detection Source	Triggering Conditions
Seat Belt	Detect if the driver is not wearing a seat belt	While the vehicle is driving, the system terminal should be able to detect that the driver is wearing a seat belt, and provide in-car and remote alarms when the driver does not wear a seat belt for 20 seconds or more.

Take your hands off the steering wheel	Detect whether the driver's hands are off the steering wheel	While the vehicle is driving, the system terminal should be able to detect the driver's hands taking off the steering wheel, and provide in-car and remote alarms when the driver's hands take off the steering wheel for 4 seconds or more.
Play Cell Phone	When the vehicle is in a driving state, it can conduct comprehensive identification and analysis based on the handheld phone items and the actions of making and receiving calls.	The system terminal should be able to identify the driver's behavior of playing with the mobile phone. When the driver continues to play with the mobile phone for 4 seconds or more, the system will provide in-car alarm and remote alarm.

5.4. BSD

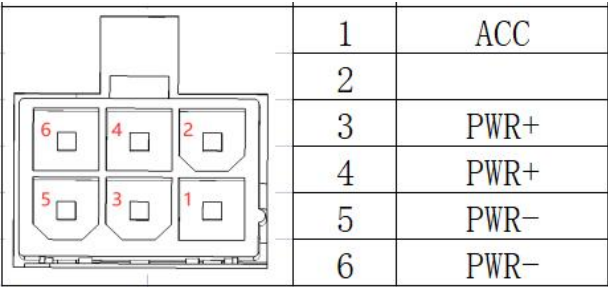
Alarm Type	Detection Source	Triggering Conditions
Front Blind	Pedestrians and non-motorized vehicles (bicycles, motorcycles, electric vehicles, tricycles and other exposed traffic participants) in front of the vehicle, including the front, side and back	Real-time detection of pedestrians and non-motor vehicles (exposed traffic participants such as bicycles, motorcycles, battery cars, tricycles, etc.) in front of the vehicle, including whether the front, side and back enter the alarm classification area (can be set during calibration)
Rear Blind	Pedestrians and non-motor vehicles (exposed traffic participants such as bicycles, motorcycles, scooters, tricycles, etc.) behind the vehicle, including the front, sides and back	Real-time detection of pedestrians and non-motor vehicles (exposed traffic participants such as bicycles, motorcycles, battery cars, tricycles, etc.) behind the vehicle, including whether the front, side and back enter the alarm classification area (can be set during calibration)
Left Blind	Pedestrians and non-motorized vehicles (exposed traffic participants such as bicycles, motorcycles, scooters, tricycles, etc.) on the left side of the vehicle, including the front, side and back	Real-time detection of pedestrians and non-motor vehicles (exposed traffic participants such as bicycles, motorcycles, battery cars, tricycles, etc.) on the left side of the vehicle, including whether the front, side and back enter the alarm classification area (can be set during calibration)
Right Blind	Pedestrians and non-motor vehicles (exposed traffic participants such as bicycles, motorcycles, scooters, tricycles, etc.) on the right side of the vehicle, including the front, sides and back	Real-time detection of pedestrians and non-motor vehicles (exposed traffic participants such as bicycles, motorcycles, battery cars, tricycles, etc.) on the right side of the vehicle, including whether the front, side and back enter the alarm classification area (can be set during calibration)

6.Interface Definition

6.1. Wire Interface Connection Methods

This paper mainly introduces the definition of power supply, I / O, audio and video interfaces, as follows:

1) Definition of power interface



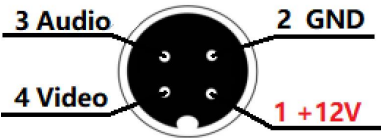
Chat 3. Definition of power interface

2) I / O interface definitions

19 Sensor_out	17 SENSOR_IN3	15 SENSOR_IN1	13 UART1_232TX	12 UART3_232RX/485A	10 GND	8 SPEAKER+	6 AOUT_MIC	4 GND	2 GND
20 ALM_IN	18 SENSOR_IN4	16 SENSOR_IN2	14 UART1_232RX	11 UART3_232TX/485B	9 GND	7 SPEAKER-	5 MIC_IN	3 12V_OUT	1 5V_OUT

Chat 4. 20PIN

3) Definition of audio and video interface



Av-in1 ~ 4 camera interface



Av-out display interface